Tunable Low-Frequency Noise in Dual-Gate MoS_2 Transistors

Xuefei Li, Tiaoyang Li, Zhenfeng Zhang, Xiong Xiong, Sichao Li, and Yanqing Wu

***Abstract*— We have systematically studied the effect of the back-gate voltage on the low-frequency noise properties of the MoS_2 transistors from 300 to 20 K in this work. The results show that the performance of the top-gate MoS_2 transistor can be effectively tuned by the back-gate voltage V_{bg}. When V_{bg} increases to 20 V, the maximum on-current increases up to 588 μA/μm for the 1-μm channel length device, as well as five times reduction of the low-frequency noise and two times reduction in contact resistance. The Fermi-level modulation by adjusting V_{bg} turns out to be an effective way of improving contact resistance and low-frequency noise for future high-quality MoS_2 metal–oxide–semiconductor field-effect transistors.**

***Index Terms*— MoS_2, dual-gate, low-frequency noise, transistors, high-*k*.**

I. Introduction

RECENTLY, molybdenum disulfide (MoS_2) has received much attention due to reasonably high mobility, high on/off ratio, and wafer-scale synthesis by chemical vapor deposition [1]–[3]. Despite the exciting progress of MoS_2 devices in numerous applications, there are still fundamental obstacles to be overcome before they can have any industrial applications [4], [5]. As one of the most important factors in electrical transport, contact resistance has been studied by several methods including external doping [6]. However, in a back-gate device, the 2D material beneath the metal contacts and in the channel region cannot be independently controlled by the back-gate voltage. As a result, it is rather phenomenological to study the contact resistance impact on device performance using such device structures. Another important factor, low-frequency noise analysis is a powerful characterization method for performance evaluation due to the non-destructive and sensitive method, and is widely used in the Si CMOS technology as listed in ITRS [7]. Furthermore, low-frequency noise determines the signal-to-noise ratio in circuits and is a major concern in radio frequency devices [8]. Although the low-frequency noise of back-gate MoS_2 field-effect transistors (FETs) has been investigated by several groups [9]–[15], the noise level is still higher than that of modern Si transistors and more importantly, the noise characteristics of dual-gate MoS_2 transistors are yet to be reported. Little is known on the noise mechanisms of dual-gate MoS_2 transistors where the contact and channel can be controlled separately. As a result, it is important in terms of both device physics and technology to understand the low-frequency noise behavior and find a reliable way to optimize it in the dual-gate MoS_2 transistors. In such a structure where the contact and channel can be modulated separately by the back-gate and top-gate voltages, the impact of contact resistance on output performance can also be clearly resolved as well as the effect on the noise magnitude. This unique device structure offers a guide for future optimizations of electrical performance of 2D materials.

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II. Experiment

The transistor fabrication began with exfoliation of MoS_2 flakes from bulk material onto a p-Si/SiO_2 substrate with a SiO_2 thickness of 90 nm, which was pre-patterned with alignment marks. Then, the sample was soaked in 1,2-dichloroethane for 12 hours. Electron-beam lithography was used to pattern source and drain contacts using PMMA, and then Ni/Au (15/45 nm) was evaporated and lifted off to form ohmic contacts. Following source and drain metallization, a dual-layer dielectric stack was deposited by atomic layer deposition (ALD) on the MoS_2 channel, which was composed of 25-nm Al_2O_3 and 20-nm HfO_2, corresponding to an equivalent oxide thickness (EOT) of 23 nm (the dielectric constant of Al_2O_3 and HfO_2 is 6 and 12, respectively). Al_2O_3 was deposited with trimethyl-aluminum and H_2O as precursors at 75 °C and then 20 nm HfO_2 was deposited in situ using tetradimethylaminohafnium and H_2O as precursors at 150 °C. Al_2O_3 is mainly used as an interfacial layer and HfO_2 as a high-*k* dielectric layer to boost the overall dielectric constant [1]. Following the ALD process, the gate electrode was defined by electron-beam lithography and 20 nm/45 nm of Ti/Au layers were electron-beam evaporated. The thickness of the MoS_2 layer is 4.3 nm measured by atomic force microscopy. Fig. 1a and 1b are device schematic image and scanning electron microscopy image of the fabricated dual-gate MoS_2 transistors, respectively. The channel length and

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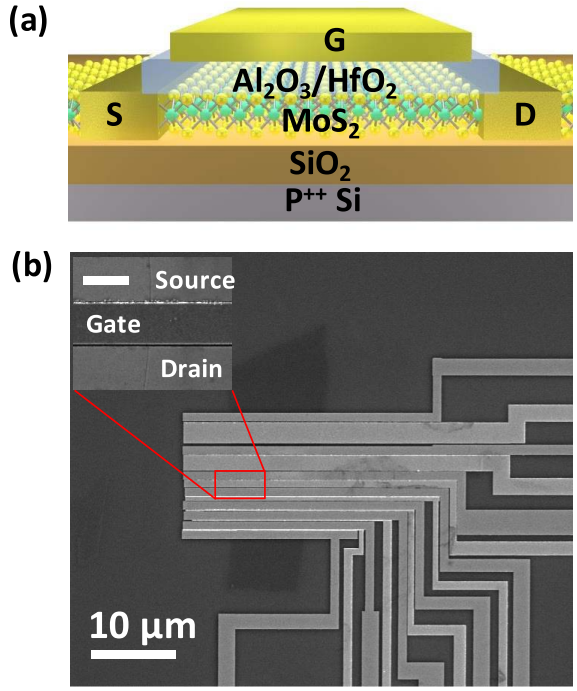


Fig. 1. (a) Schematic view of dual gate MoS_2 transistors. (b) SEM image of a MoS_2 dual-gated FET. The inset in (b) shows the zoom-in image of the top-gate MoS_2 transistor with 1 μm channel length. Scale bar of the inset of (b), 1 μm.

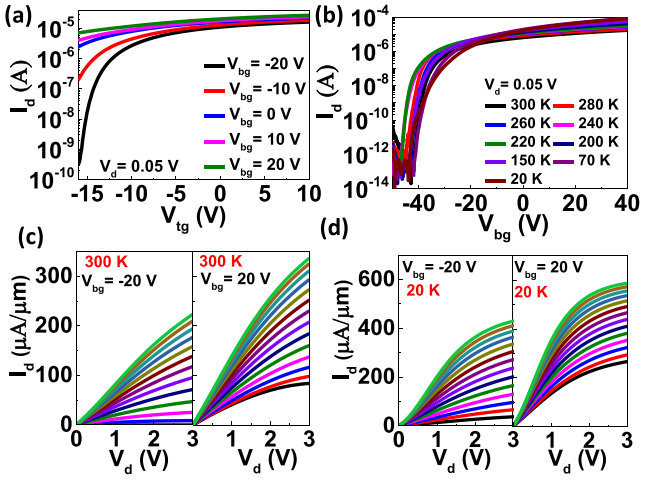


Fig. 2. (a) Transfer characteristics of the MoS_2 top-gate FET at V_d = 0.05 V for five different V_{bg} at room temperature. (b) The transfer characteristics of the MoS_2 back-gate FET at V_d = 0.05 V from 300 to 20 K. (c)-(d) Output characteristics of MoS_2 dual-gate MOSFET with top-gate voltage from −16 to 10 V while fixing $V_{bg} = -20$ and 20 V at 300 and 20 K, respectively.

width of the MoS_2 transistor is 1 and 9.2 μm, respectively. All the electrical measurements were carried out under vacuum ($<10^{-5}$ Torr) in a Lakeshore cryogenic probe station.

III. Results and Discussion

Fig. 2a shows the drain current I_d versus top-gate voltage V_{tg}, at a drain voltage of 0.05 V for the MoS_2 FET with V_{bg} changing from −20 V to 20 V at 300 K. It can be seen that the off current is very small and the I_{on}/I_{off} ratio is over 10^5 when V_{bg} is at −20 V. The hysteresis is ∼0.6 V when V_{tg} was swept from −16 V to 10 V and back to −16 V, which could be mainly attributed to the charge traps at interface and in the dielectric films. The sub-threshold swing of the top-gate MoS_2 device from the forward and backward sweep at $V_d = 1$ V are 390 and 440 mV/dec, respectively. With increasing V_{bg}, the threshold voltage shifts toward negative voltage and the I_{on}/I_{off} ratio decreases accordingly, which could be attributed to the enhanced doping concentration at the contacts by the back gate and reduced electrostatic control of the multilayer MoS_2 channel by the top gate, similar to the previous studies by other groups [6], [16]. Meanwhile, it should be noted that the drain current at $V_d = 0.05$ V increases monotonically with V_{bg} from −20 to 20 V due to the increased carrier density in both channel region and the contact regions. Fig. 2b shows the transfer characteristics of the dual-gate MoS_2 device at V_d = 0.05 V with V_{bg} varying from −50 to 40 V and $V_{tg} = 0$ V from 300 to 20 K. An I_{on}/I_{off} ratio greater than 10^8 is obtained at all the temperatures owing to its relatively large bandgap. Fig. 2c shows a comparison of the output characteristics of the dual-gate device for two different V_{bg} at 300 K. The top-gate bias V_{tg} varies from −16 to 10 V with a 2 V step. The maximum I_d at $V_d = 3$ V with $V_{bg} = -20$ and 20 V is 220 and 337 μA/μm, respectively. The 53% increase in the maximum on-current is comparable with that of PEI doped MoS_2 FETs, which is caused by the reduction of contact resistance R_c and sheet resistance in the channel [6]. Similar behavior can be obtained at low temperature of 20 K as shown in Fig. 2d. When V_{bg} increases to 20 V, the maximum on-current increases up to 588 μA/μm from 427 μA/μm at $V_{bg} = -20$ V. Larger V_{bg} induces more carriers in the MoS_2 layer beneath the contact metals, resulting in a very narrow Schottky barrier and small R_c. Moreover, the carrier transport at the source-end junction can also benefit from the increased electrostatic doping of MoS_2 in both of the contact region and channel region, similar to the graphene-metal junction by back-gate induced doping [17].

To gain further insight into the role of V_{bg} on the contact region, we performed temperature-dependent measurement to evaluate the impact on R_c. R_c is extracted using the transfer length method and the V_{tg} is fixed at the threshold voltage (V_{th}). Fig. 3a shows the R_c as a function of temperature at various V_{bg} for the same device. The contact resistance R_c decreases with increasing V_{bg} in the whole temperature range, which demonstrates that the contact resistance R_c can be tuned by the V_{bg}. The R_c for $V_{bg} = -20$ and 20 V is 3.9 and 1.8 k$\Omega \cdot \mu$m, respectively. In addition, the R_c also decreases with decreasing temperature, indicating that the carrier transport is more dominated by tunneling instead of thermionic emission. Our previous work also demonstrated the same mechanism, similar to the work using graphene as the contact [18]. Fig. 3b shows the contour plot of I_d as a function of V_{tg} and V_{bg} at 300 K with V_d fixed at 0.2 V. It is found that I_d depends on the values of V_{bg} as well as V_{tg}. It is clear that the drain current I_d increases with V_{tg} and V_{bg} with local linear contours. Fig. 3b shows the transfer characteristics of the top-gate MoS_2 FET at $V_d = 0.2$ V with V_{bg} fixed at different values. When V_{bg} is larger, the transfer curve moves towards more negative voltages, indicating a n-type doping effect of the V_{bg}. Based on the data in Fig. 3c, we can extract

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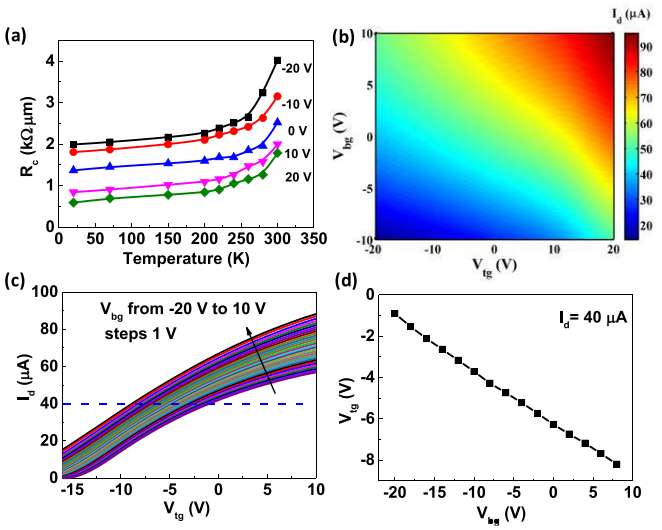


Fig. 3. (a) The contact resistance R_c as a function of temperature at different Vbg. (b) Contour plot of drain current I_d (μA) as function of V_{bg} and V_{tg} with $V_d = 0.2$ V at room temperature. (c) I_d-V_{tg} characteristics of MoS_2 dual-gate MOSFET with top-gate voltage stepped from -16 to 10 V with $V_d = 0.2$ V while changing V_{bg} from -20 to 10 V with a 1 V step at 300 K. (d) The linear relation between V_{bg} and V_{tg} at 300 K.

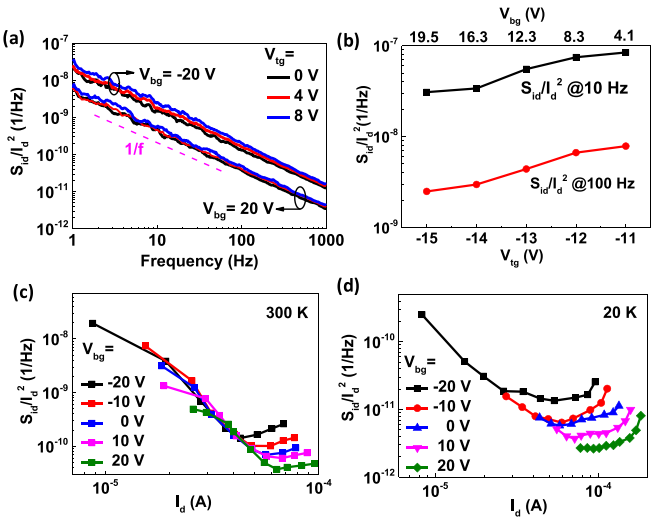


Fig. 4. (a) Normalized noise spectral density (S_{id}/I_d^2) as a function of frequency with $V_d = 0.2$ V at three different top-gate voltages for $V_{bg} = -20$ and 20 V at 300 K. The ideal $1/f$ behavior is added for comparison. A clear $1/f$ characteristic is observed. (b) Normalized noise spectral density (S_{id}/I_d^2) at two different frequencies for five sets of different V_{bg} and V_{tg}. The drain current is fixed at 17 μA. (c)-(d) Normalized noise spectral density (S_{id}/I_d^2) at $f = 100$ Hz as a function of I_d at various V_{bg} with $V_d = 0.2$ V for 300 and 20 K, respectively.

a set of values of V_{tg} and V_{bg} for $I_d = 40$ μA as shown in Fig. 3d. V_{bg} is linear with V_{tg} and the slope is $\sim$0.26, consistent with the expected value of $EOT_{top-gate}/EOT_{back-gate}$, where $EOT_{back} = 90$ nm for the back-gate SiO_2, and $EOT_{top} = 23$ nm for the Al_2O_3/HfO_2 top-gate dielectric stack.

To investigate the effect of V_{bg} on the noise behavior, the noise measurements were performed with V_{tg} swept from -16 to 10 V and V_{bg} fixed at -20, -10, 0, 10, and 20 V. Fig. 4a shows a comparison of the normalized current noise spectral density (S_{id}/I_d^2) of the top-gate MoS_2 FET at $V_{bg} = -20$ and 20 V with three different top-gate biases in the 1-1000 Hz range at $V_d = 0.2$ V. A typical $1/f^{\gamma}$ behavior with γ close to unity can be observed for all the curves. One important observation is that the normalized drain-current noise magnitude for $V_{bg} = 20$ V is significantly smaller than the other V_{bg} due to the improved contact resistance. To further verify the role of contact resistance on the noise magnitude, we changed the V_{bg} and V_{tg} on the opposite direction while fixing $I_d = 17$ μA (with variation less than 5%) and then performed the noise measurement at five different sets of V_{tg} and V_{bg}. Fig. 4b shows the drain current noise extracted at $f = 10$ and 100 Hz at room temperature. At both frequencies, the noise magnitude of the top-gate MoS_2 transistor decreases with larger V_{bg}. A nearly 3 times reduction of noise magnitude can be achieved when V_{bg} increases from 4.1 to 19.5 V, indicating that the noise level of the top-gate MoS_2 transistor can be effectively tuned by the V_{bg}. The normalized current noise spectral density S_{id}/I_d^2 extracted at the frequency of 100 Hz as a function of drain current I_d with five different V_{bg} at 300 K is shown in Fig. 4c. The noise behavior shows the similar trend on drain current for all V_{bg}, indicating the same physical mechanism. In the strong inversion region, the noise level decreases by a factor of 5 with increased V_{bg}. This shows that the contact resistance is the dominating contribution to the transistor noise [19], which is in agreement with previous studies that found that contact resistance plays an important role in high drain current region [9]. However, at small drain current regime, the slope of S_{id}/I_d^2 curves is almost independent of V_{bg}, which implies that the main source of noise is the transistor channel. Fig. 4d shows the S_{id}/I_d^2 versus I_d for five different V_{bg} at 20 K. It is clear that the noise first decreases with drain current, and then starts to increase when the I_d is larger than 100 μA, which could be attributed to the dominant effect from the contact resistance [11], [20]. Meanwhile, the effect of interface traps could also play an important role in the strong inversion region, which will be systematically studied in future work. With larger V_{bg}, the S_{id}/I_d^2 curve shifts toward the positive voltage and reduces monotonically due to the decrease in contact resistance and improved drain current. The results show that improvement in contact resistance is crucial to reduce the noise level, which is important for high-performance devices.

IV. Conclusion

In conclusion, we have demonstrated that contact resistance plays a critical role in the performance of the MoS_2 device using a dual-gate structure. We have also demonstrated effective control of the low-frequency noise in 2D device using the electrostatic doping by a back-gate voltage. A 53% improvement in drain current, 2 times decrease in contact resistance, and 5 times reduction in noise level in the top-gate MoS_2 transistors can be achieved by optimizing device operation through a back-gate voltage. The results reported here provide a new path to explore high-performance 2D device and promote further applications including electronics, photonics and sensors.

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